

• Learn E . t. $d(\phi, \phi') \triangleq ||E(\phi) - E(\phi')||_2$ is a bisimulation metric, which gives us for all ϕ, ϕ' :

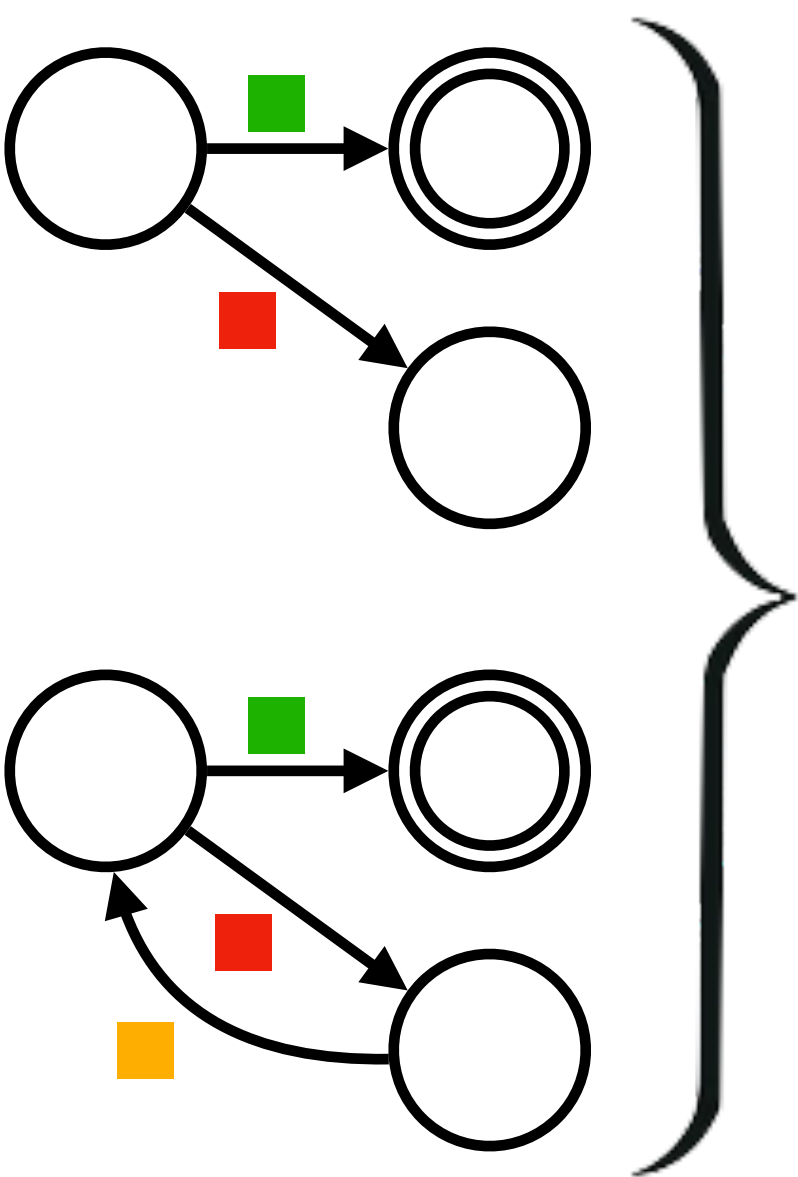
$$d(\phi, \phi') = 0 \iff E(\phi) = E(\phi') \iff \phi \sim \phi'.$$

A encoder E takes inputs and outputs embeddings, trained by learning a bisimulation metric

Two automata ϕ and ϕ' are called bisimilar if they represent the same task, denoted by $\phi \sim \phi'$.

RAP Embeddings

d	ϕ_1	ϕ_2	ϕ_3	$\dots$
ϕ_1	0	a	b	$\dots$
ϕ_2	a	0	c	$\dots$
ϕ_3	b	c	0	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

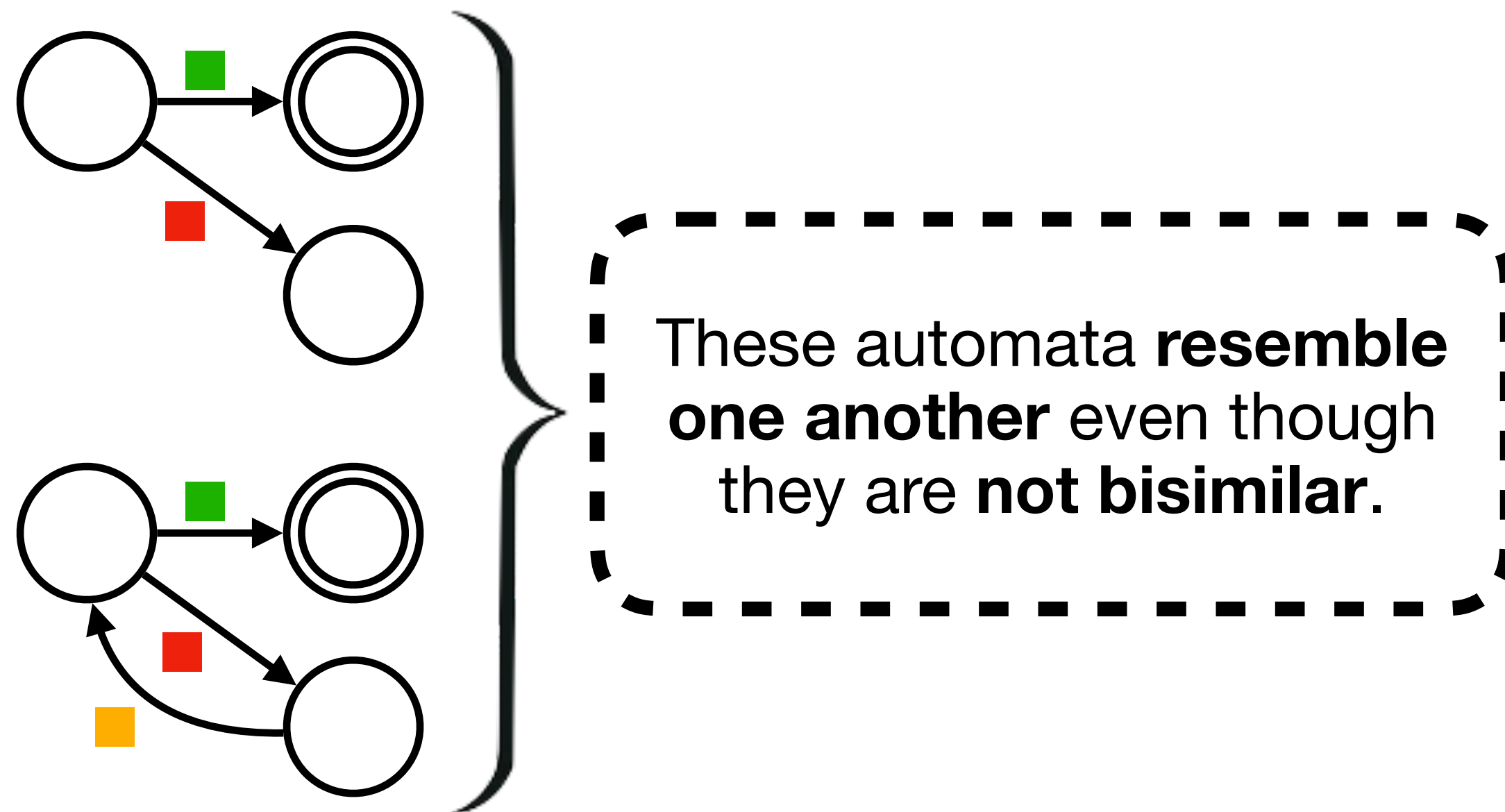


These automata **resemble one another** even though they are **not bisimilar**.



RAD Embeddings

- A encoder E takes **automata** and outputs **embeddings**, trained by learning a **bisimulation metric**.
- Two **automata** ϕ and ϕ' are called **bisimilar** if they **represent the same task**, denoted by $\phi \sim \phi'$.



d	ϕ_1	ϕ_2	ϕ_3	$\dots$
ϕ_1	0	a	b	$\dots$
ϕ_2	a	0	c	$\dots$
ϕ_3	b	c	0	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

- Learn E s.t. $d(\phi, \phi') \triangleq \|E(\phi) - E(\phi')\|_2$ is a **bisimulation metric**, which gives us for all ϕ, ϕ' :

$$d(\phi, \phi') = 0 \iff E(\phi) = E(\phi') \iff \phi \sim \phi'.$$

***RAD** Embeddings*

- Improve sample efficiency in TC-RL and TC-MARL.